

Deba Priyo Guha

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CURRENT POSITION

Full-time Research Assistant, Center for Astronomy, Space Science and Astrophysics (CASSA), Independent University, Bangladesh (IUB) — since February 2026. Working on Type Ia supernova cosmology under Dr. Syed Ashraf Uddin.

RESEARCH INTERESTS

Machine Learning and Deep Learning for Astrophysics, Type Ia Supernova Cosmology and the Hubble Tension, Radio Galaxy Morphology (AGN), Astronomical Transient Classification, Interpretable and Physics-Informed AI, Time-Domain Astrophysics, Computer Vision for Scientific Imaging, Multimodal and Vision Language Models.

EDUCATION

Rajshahi University of Engineering and Technology (RUET)

Feb 2019 – Sept 2024

B.Sc. Electrical and Computer Engineering CGPA: 2.88/4.00

Thesis: *PIE-ADA: Physics-Informed Ensemble with Adaptive Data Augmentation for Photometric Transient Classification* — Supervisor: Fariya Tabassum

HSC (Govt. City College Chattogram, 2018) GPA 4.83/5.00 **SSC** (Nasirabad Govt. High School, 2016) GPA 5.00/5.00 (perfect)

IELTS Academic (British Council, Sept 2025): Overall **7.0** Listening 8.0 | Reading 7.5 | Writing 6.5 | Speaking 6.0

RESEARCH EXPERIENCE

Research Assistant (Full-time), CASSA, Independent University Bangladesh

Feb 2026 – Present

Supervisor: Dr. Syed Ashraf Uddin (Associate Professor, CASSA)

- Investigating host galaxy effects on Type Ia Supernova luminosity to address the Hubble tension (5σ discrepancy in H_0)
- Building Bayesian SED-fitting and nested-sampling pipelines (BAGPIPES, Nautilus) for host galaxy age analysis and light-curve calibration of SNe Ia

Research Intern, CASSA, Independent University Bangladesh

Oct 2025 – Present

Supervisor: Dr. Khan Muhammad Bin Asad (Director, CASSA)

- Contributing to **RGC v2.0** (Radha Gobinda Chandra), an open-source Python package for automated radio AGN morphology classification using 20,000+ LOFAR/LoTSS and VLA/FIRST images
- Implementing semi-supervised CNN (BYOL + E2CNN) for Wide-Angle Tail / Narrow-Angle Tail classification; developing spurious source filtering algorithms
- Curated the dataset and catalogue for **RGC v1.0** (accepted, *A&A* 2026; arXiv:2510.22190); building a Vision-LLM pipeline (CLIP, BLIP) for multi-wavelength galaxy characterization

Research Collaborator, Young Learners' Research Lab, CSE, RUET

2024 – Present

Supervisor: Md. Azmain Yakin Srizon (Assistant Professor, CSE)

- Leading **LunaXNet**: a dual-path framework (classical PCA/NMF/GMM versus a Spectral Swin-UNet in a shared CVAE latent space) for lunar mineral mapping from Chandrayaan-1 M3 hyperspectral data (73 usable bands); band-parameter mineralogy validated on withheld Apollo 12/15 sites, with Spectral Grad-CAM and Monte Carlo Dropout for explainability and per-pixel uncertainty
- Co-authoring research on GAN-enhanced gastrointestinal disease classification and BERT/GPT-2 based spam detection

Undergraduate Thesis Researcher, ECE, RUET

Jan 2024 – Sept 2024

Supervisor: Fariya Tabassum

- Developed **PIE-ADA**: 14-class photometric transient classifier on PLAsTiCC dataset; extracted 271 multi-scale features from 6 photometric passbands; astrophysically constrained augmentation (time dilation, dust extinction, correlated noise)
- Achieved weighted log-loss 0.5763 (± 0.0083), 80.33% accuracy (5-fold CV); 24–49% improvement over baselines; classifies one object in <0.05 s for real-time LSST processing

PUBLICATIONS AND MANUSCRIPTS

- [1] M. S. Hossain, M. S. H. Shahal, K. M. B. Asad, P. Saikia, A. Khan, F. Akter, A. A. Ali, M. A. Amin, **D. P. Guha**, et al., “RGC: A Radio AGN Classifier Based on Deep Learning. I. A Semi-Supervised Multi-Class Model for VLA Images,” *Astronomy & Astrophysics*, accepted (17 July 2026). DOI:10.1051/0004-6361/202557828; arXiv:2510.22190
- [2] **D. P. Guha**, F. Tabassum, “PIE-ADA: Physics-Informed Ensemble with Adaptive Data Augmentation for Photometric Transient Classification,” *2026 IEEE 2nd Int. Conf. on Quantum Photonics, AI & Networking (QPAIN 2026)*, IEEE. Published. DOI:10.1109/QPAIN69676.2026.11546175; arXiv:2606.29367. [First Author]
- [3] **D. P. Guha**, M. A. Y. Srizon, et al., “LunaXNet: A Dual-Path Hyperspectral AI Framework for Lunar Mineral Mapping with Explainability and Uncertainty Estimation,” *Target: IEEE Transactions on Geoscience and Remote Sensing*. [First Author]
- [4] M. S. H. Shahal, A. Khan, **D. P. Guha**, K. M. B. Asad, et al., “RGC v2.0: Hierarchical Multi-Class Classification of Radio AGN Morphologies,” *Target: A&A*.
- [5] M. S. Hossain, **D. P. Guha**, K. M. B. Asad, et al., “Vision-Language Models for Multi-Wavelength Galaxy Characterization,” *Target: ApJS*.
- [6] T. Sabira, **D. P. Guha**, M. A. Y. Srizon, et al., “GAN-Enhanced Gastrointestinal Disease Classification,” *Target: Medical Image Analysis*, Elsevier.
- [7] S. Rezwana, **D. P. Guha**, “Physics-Inspired Multimodal Emotion Recognition for Telemedicine Triage,” *Target: upcoming IEEE conference (2027)*. In preparation.

TECHNICAL SKILLS

Languages: Python (5+ yrs), C/C++ (4+ yrs), MATLAB (3+ yrs); SQL, PHP, JavaScript, Bash

ML/DL: PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, LightGBM; CNN, LSTM, ViT, BERT, GPT-2, CLIP, BLIP, EfficientNet, StyleGAN2-ADA; SHAP, Grad-CAM, physics-informed learning

Astronomy: Astropy, Astroquery, APLpy, TOPCAT, DS9, BAGPIPES, Nautilus; LOFAR/LoTSS, VLA/FIRST, Pan-STARRS, SDSS, WISE; FITS, hyperspectral analysis, radio interferometry

Tools: Git/GitHub, Linux/Unix, Jupyter, HPC (SLURM), Google Colab; Flask, MySQL, Arduino, Raspberry Pi

SELECTED AWARDS AND HONORS

- **IAU Hands-on Workshop in Radio Astronomy & 10th Byurakan International Summer School** (BAO, Armenia, Sept 2026) — selected among 39 participants from 19 countries
- **Bangladesh National Team, IOAA 2017** (11th International Olympiad on Astronomy & Astrophysics, Phuket) — selected as 1 of 5 national representatives
- **Silver Medal** (2023, 2020) and **Accomplished Winner** (2022) — University Physics Competition (International)
- **Bronze Honor** — International Astronomy & Astrophysics Competition (IAAC, 2021) and International Youth Math Challenge (IYMC, 2021)
- **UGC Technical Scholarship**, University Grants Commission, Bangladesh (3 consecutive years, 2020–2022)
- **National Winner**, 12th Bangladesh Astro-Olympiad (2017); **Regional Champion**, Bangladesh Physics Olympiad (2018)

LEADERSHIP AND SERVICE

President, Astronomy and Science Society of RUET (ASSR)

June 2022 – May 2023

- Led 50+ member organization; organized **AURORA: National Science Festival 2023** (500+ participants from universities across Bangladesh); served as Problem Setters’ Team Leader and Olympiad Judge

Academic Problem Setter and Mentor

2019 – Present

Bangladesh Mathematical Olympiad (BdMO), Bangladesh Physics Olympiad (BdPhO), and BDOAA — designing problems for regional and national rounds; mentoring students for international olympiads

Founder and Lead Instructor, STEM by Priyojeet, Chattogram

Dec 2018 – Present

- Mentored 18+ students in physics, mathematics, and programming; students gained admission to BUET, RUET, CUET, and medical colleges

REFERENCES

Dr. Syed Ashraf Uddin

Associate Professor, CASSA, IUB

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Md. Azmain Yakin Srizon

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Dr. Khan Muhammad Bin Asad

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Fariya Tabassum

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